# Impact of Euler Angle Singularities on UAV Attitude Control: A Comparison with Quaternion-Based Formulations

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**Abstract.** Accurate attitude representation is critical for unmanned aerial vehicle (UAV) modeling and control, especially in safety-critical missions involving large attitude variations. While Euler-angle representations are widely used due to their simplicity, they suffer from kinematic singularities that can degrade control performance near critical configurations.

This paper investigates the impact of Euler-angle singularities on UAV attitude control by comparing Euler-angle and quaternion-based formulations under identical proportional–derivative control laws. Numerical simulations explicitly drive the system toward and across singular configurations to isolate representation-induced effects.

The results show that Euler-angle formulations exhibit a sharp increase in angular velocity demand and control effort near singularities, whereas quaternion-based formulations maintain smooth and stable attitude evolution. These findings highlight the limitations of Euler-angle representations and support the use of quaternions for robust UAV attitude control.

**Keywords:** UAVs, attitude representation, Euler angles, quaternions, numerical stability

## 1 Introduction

Unmanned Aerial Vehicles (UAVs) are widely employed in civil and industrial applications such as surveillance, mapping, infrastructure monitoring, and autonomous inspection. Among these, autonomous power transmission line inspection poses particularly stringent requirements on navigation accuracy, attitude-control robustness, and numerical reliability, due to close-proximity operation and large attitude variations during inspection maneuvers [7].

In such missions, the choice of attitude representation plays a fundamental role in flight safety and control performance. Euler-angle parametrizations

are still commonly adopted in simulation environments and control prototyping because of their intuitive interpretation and low computational cost. However, Euler angles suffer from inherent kinematic singularities, commonly referred to as gimbal lock, which occur when the pitch angle approaches $\pm\frac{\pi}{2}$. Near these configurations, the mapping between body angular velocity and attitude rates becomes ill-conditioned, potentially leading to numerical instability and degraded control performance [6].

Quaternion-based representations provide a globally nonsingular description of rigid-body orientation, ensuring smooth and continuous attitude evolution over the entire configuration space. For this reason, quaternion formulations are employed in modern UAV flight control systems, including Pixhawk/PX4-based platforms [8]. Nevertheless, in simulation-based studies and simplified control designs, the practical consequences of Euler-angle singularities are often underestimated or overlooked.

This work investigates how the choice of attitude representation influences UAV control behavior in reliability-critical applications. By comparing Euler-angle and quaternion-based formulations under identical control laws, numerical simulations are used to isolate the structural effects of attitude parametrization. The results show that Euler-angle formulations exhibit a pronounced increase in angular-rate demand and control effort as the system approaches singular configurations, whereas quaternion-based formulations maintain well-conditioned and stable behavior. These effects arise from the mathematical structure of the representation itself, reinforcing the suitability of quaternion-based approaches for robust UAV attitude control [4].

### 1.1 Objectives

The objective of this work is to analyze the influence of attitude representation on UAV control performance, with particular emphasis on the kinematic singularities associated with Euler-angle formulations.

Specifically, this study aims to:

- compare Euler-angle and quaternion-based attitude representations under identical control laws;
- evaluate system behavior near and across singular configurations;
- assess attitude response, angular velocity demand, and control effort;
- discuss the implications of each representation for reliability-critical UAV applications.

The theoretical background and problem formulation underlying these comparisons are introduced in the next section.

## 2 Background and Problem Formulation

The orientation of a rigid body evolves on the special orthogonal group $SO(3)$, and any attitude representation used for modeling and control can be interpreted

as a parametrization of this geometric space. The mathematical properties of such parametrizations directly affect the numerical conditioning, stability, and robustness of attitude estimation and control algorithms [4].

Euler angles are widely used due to their intuitive roll–pitch–yaw interpretation. However, they provide only a local parametrization of $SO(3)$ and exhibit kinematic singularities, commonly referred to as gimbal lock, which occur when the Jacobian relating angular velocity to attitude rates loses rank, typically at pitch angles $\theta = \pm\frac{\pi}{2}$. Near these configurations, the kinematic equations become ill-conditioned, leading to high numerical sensitivity and degraded control performance [5].

Unit quaternions offer a globally nonsingular alternative by representing rotations on the unit three-sphere $S^3$. Despite introducing redundancy through the unit-norm constraint, quaternion formulations preserve a smooth and well-conditioned relationship between angular velocity and attitude evolution over the entire configuration space, avoiding coordinate singularities and numerical degradation under large attitude variations [3].

From a control perspective, the attitude representation has a direct impact on closed-loop behavior. Euler-angle formulations may induce excessive control effort or numerical instability near singular configurations, whereas quaternion-based formulations maintain smooth attitude dynamics and numerical robustness. This makes quaternions particularly suitable for aggressive maneuvers and safety-critical UAV applications [2].

In autonomous UAV missions involving large attitude excursions and high reliability requirements, such as power transmission line inspection, representation-induced numerical issues become critical. This motivates the comparative framework presented in the next section, where Euler-angle and quaternion-based formulations are evaluated under identical modeling and control conditions.

## 3 Modeling, Control Design, and Simulation Setup

This section presents the modeling framework, attitude control design, and numerical setup adopted in this study. The objective is to establish a consistent and transparent formulation that enables a fair comparison between Euler-angle and quaternion-based attitude representations under identical control conditions, without introducing additional modeling or control complexities that could obscure representation-induced effects [5].

All simulations were performed in discrete time using a fixed integration step. A simplified quadrotor model was adopted, with emphasis on attitude dynamics and control effort rather than high-fidelity aerodynamic effects. A proportional–derivative (PD) controller with identical gain values was employed in both formulations. External disturbances, sensor noise, and unmodeled dynamics were intentionally excluded in order to isolate the effects of attitude parametrization. The reference commands were designed to drive the system toward and across the Euler kinematic singularity.

### 3.1 Attitude Kinematics

The UAV is modeled as a rigid body whose orientation evolves on the special orthogonal group $SO(3)$. Two attitude parametrizations are considered: Euler angles and unit quaternions.

In the Euler-angle formulation, the vehicle attitude is described by the roll, pitch, and yaw angles,

$$\boldsymbol{\eta} = [\phi,\ \theta,\ \psi]^T, \tag{1}$$

and the kinematic relationship between Euler angle rates and the body angular velocity $\boldsymbol{\omega} = [p,\ q,\ r]^T$ is given by

$$\dot{\boldsymbol{\eta}} = \mathbf{T}(\boldsymbol{\eta})\boldsymbol{\omega}, \tag{2}$$

where $\mathbf{T}(\boldsymbol{\eta})$ denotes the Euler kinematic transformation matrix. This matrix becomes singular when the pitch angle satisfies $\theta = \pm\frac{\pi}{2}$, leading to severe numerical ill-conditioning in the mapping between angular velocity and attitude rates [7].

In the quaternion-based formulation, the attitude is represented by the unit quaternion

$$\mathbf{q} = [q_0,\ q_1,\ q_2,\ q_3]^T, \qquad \|\mathbf{q}\| = 1, \tag{3}$$

whose kinematics are governed by

$$\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega})\mathbf{q}, \tag{4}$$

where $\boldsymbol{\Omega}(\boldsymbol{\omega})$ is the skew-symmetric matrix constructed from $\boldsymbol{\omega}$. Unlike the Euler-angle formulation, the quaternion kinematics are globally nonsingular and remain well conditioned over the entire configuration space [7].

### 3.2 Attitude Control Law

To ensure an unbiased comparison between the two attitude representations, the same proportional–derivative (PD) control structure and gain values are employed in both cases.

For the Euler-angle formulation, the attitude control law is defined as

$$\boldsymbol{\omega}_c = K_p(\boldsymbol{\eta}_{ref} - \boldsymbol{\eta}) - K_d\dot{\boldsymbol{\eta}}, \tag{5}$$

where $K_p$ and $K_d$ are positive-definite gain matrices. Due to the Euler kinematics in (2), the resulting angular velocity command $\boldsymbol{\omega}_c$ is indirectly affected by the conditioning of $\mathbf{T}(\boldsymbol{\eta})$, leading to excessive angular-rate demands as the system approaches the singular configuration [7].

For the quaternion-based formulation, the attitude error is defined as

$$\mathbf{q}_e = \mathbf{q}_{ref}^{-1} \otimes \mathbf{q}, \tag{6}$$

and the control law is given by

$$\boldsymbol{\omega}_c = -K_p\mathbf{q}_{e,v} - K_d\boldsymbol{\omega}, \tag{7}$$

where $\mathbf{q}_{e,v}$ denotes the vector part of the quaternion error. This formulation avoids kinematic singularities and preserves smooth closed-loop attitude dynamics [1].

The modeling and control framework presented in this section provides a controlled environment in which the effects of attitude representation can be isolated. This setup serves as the basis for the numerical experiments discussed in the following section, where the impact of kinematic singularities on attitude tracking and control effort is systematically evaluated.

## 4 Simulation Results and Discussion

This section presents the numerical results obtained from the simulations, focusing on the impact of attitude representation on UAV control performance. Euler-angle and quaternion-based formulations are evaluated under identical proportional–derivative control laws, with emphasis on attitude tracking behavior, singularity-crossing scenarios, and control effort.

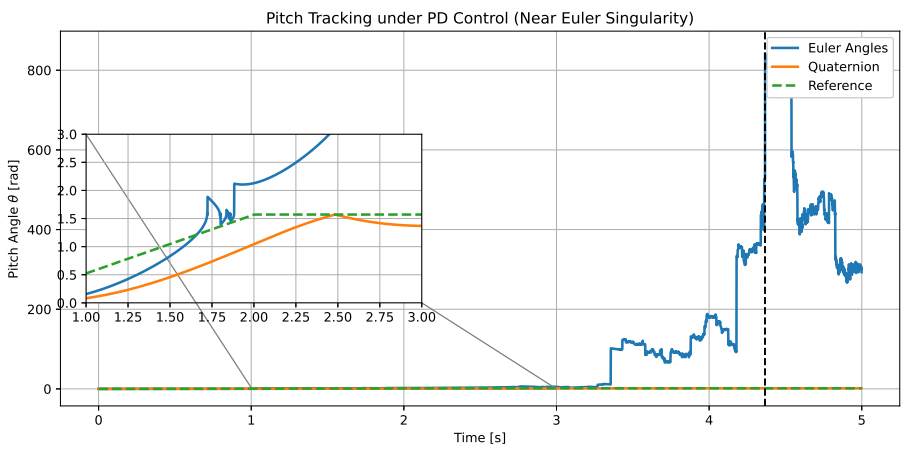


**Fig. 1.** Pitch angle tracking under PD control near the Euler kinematic singularity. While both representations behave similarly away from critical configurations, the Euler-angle formulation exhibits oscillations and numerical degradation as $\theta \approx \pm\frac{\pi}{2}$, whereas the quaternion-based formulation remains smooth and well behaved throughout the maneuver.

Figure 1 shows the pitch angle response as the reference command drives the system toward the Euler kinematic singularity. For moderate pitch angles, both representations achieve comparable tracking performance. However, as the singular configuration is approached, the Euler-angle formulation exhibits oscillatory behavior and loss of smoothness, indicating numerical degradation caused by the

ill-conditioned mapping between angular velocity and attitude rates. In contrast, the quaternion-based formulation maintains a continuous and well-behaved response, demonstrating robustness to large attitude variations.

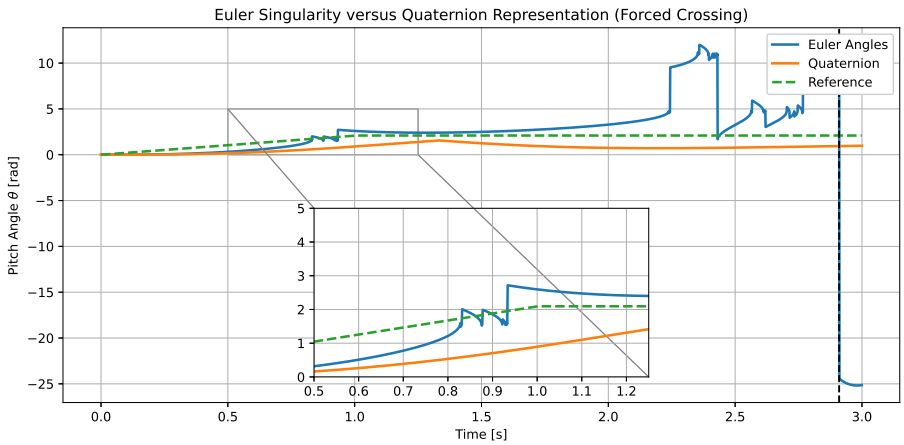


**Fig. 2.** Comparison of Euler-angle and quaternion-based attitude representations during a forced singularity-crossing maneuver. When the pitch reference exceeds $\pm\frac{\pi}{2}$, the Euler-angle formulation exhibits numerical divergence due to kinematic singularity, whereas the quaternion-based representation preserves continuous and stable attitude evolution.

To further highlight the structural limitations of Euler-angle representations, a forced singularity-crossing maneuver was performed, in which the pitch reference explicitly exceeds $\pm\frac{\pi}{2}$. As shown in Figure 2, the Euler-angle formulation undergoes numerical divergence and non-physical behavior when crossing the singular configuration. This loss of stability reflects the inability of the Euler parametrization to provide a consistent attitude description beyond the singularity. Conversely, the quaternion-based formulation preserves continuity and stability throughout the maneuver, confirming its global validity.

Figure 3 compares the angular rate demand generated by the PD controller during the singularity-crossing maneuver. As the Euler-angle formulation approaches the singular configuration, the required control effort grows rapidly, reaching values several orders of magnitude higher than those obtained with the quaternion-based formulation. This behavior is a direct consequence of the ill-conditioned Euler kinematics, whereby small attitude errors lead to disproportionately large angular velocity commands.

In contrast, the quaternion-based formulation maintains bounded and well-conditioned angular rates under identical control gains and reference commands. These results demonstrate that the observed degradation in the Euler-angle for-

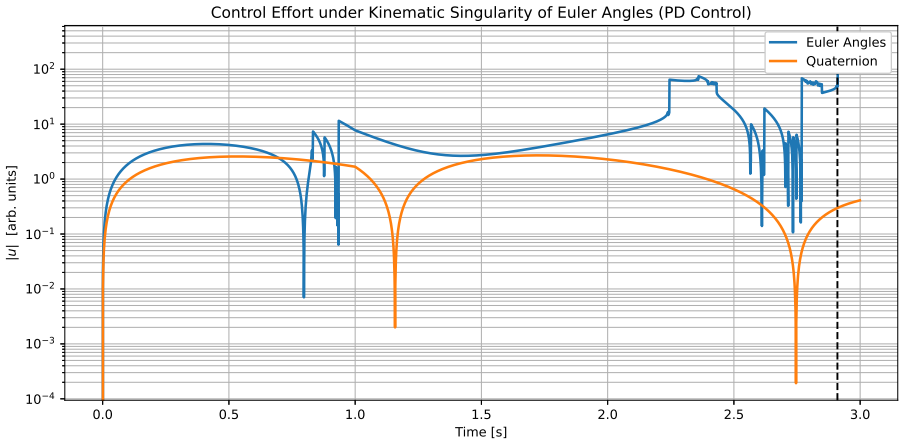


**Fig. 3.** Angular rate demand during the singularity-crossing maneuver. As the Euler-angle representation approaches the kinematic singularity, the required control effort increases sharply due to ill-conditioned kinematics, whereas the quaternion-based formulation maintains bounded angular rates under the same control gains.

mulation is not related to controller design, but rather to the mathematical structure of the attitude parametrization itself.

Overall, the results confirm that Euler-angle representations become unsuitable near kinematic singularities due to numerical instability and excessive control effort, even under simple control architectures. Quaternion-based formulations, on the other hand, remain globally nonsingular and ensure stable attitude evolution across the entire configuration space. These findings are particularly relevant for safety-critical UAV missions involving large attitude variations, such as autonomous power transmission line inspection, and provide practical justification for the use of quaternion-based representations in modern flight control systems.

## 5 Conclusion

This paper analyzed the impact of attitude representation on UAV control performance, with emphasis on the kinematic singularities inherent to Euler-angle formulations. A comparative study between Euler-angle and quaternion-based models was conducted under identical proportional–derivative control laws, targeting operating conditions relevant to safety-critical UAV missions.

The numerical results showed that, although both representations perform similarly under nominal conditions, the Euler-angle formulation undergoes pronounced degradation as the system approaches kinematic singularities. This degradation is characterized by oscillatory behavior, numerical instability, and a sharp increase in angular velocity demand and control effort. These effects

were shown to stem directly from the mathematical structure of the Euler-angle parametrization, rather than from controller design or tuning.

In contrast, the quaternion-based formulation remained numerically well conditioned, ensuring smooth attitude evolution and bounded control effort even during forced singularity-crossing maneuvers. This robustness confirms the suitability of quaternion representations for UAV applications involving large attitude variations and stringent reliability requirements.

Overall, the results provide clear theoretical and practical justification for the adoption of quaternion-based attitude representations in modern UAV flight control systems, including Pixhawk-based platforms. Future work may extend this analysis to incorporate full nonlinear dynamics, disturbance rejection, and experimental validation on real aerial vehicles.

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